

Part A: Introduction to Probability

Question 1

1981

In a game of chess between two players X , Y , of about equal ability, the player with the White pieces, having the first move, has a probability of 0.5 of winning, and the probability that the player with the Black pieces wins is 0.3. What is the probability that the game ends in a draw (i.e., neither player wins)?

The two players X , Y , play each other twice in a chess competition, each player having the White pieces once. In the competition a player who wins a game scores 1 point, a player who loses a game scores 0 points, and a draw scores as $\frac{1}{2}$ point to each player. What is the probability that, as a result of these two games,

- (a) X scores 2 points?
 (b) X scores less than $1\frac{1}{2}$ points?

$$P(\text{Draw}) = 1 - (0.5 + 0.3) \\ = 0.2.$$

Assuming player X has the white pieces in game one:

$$P(X \text{ wins game 1}) = 0.5$$

$$P(\text{Drawn game 1}) = 0.2$$

$$P(X \text{ loses game 1}) = 0.3$$

$$P(X \text{ wins game 2}) = 0.3$$

$$P(\text{Drawn game 2}) = 0.2$$

$$P(X \text{ loses game 2}) = 0.5$$

$$(a) \quad P(X \text{ scores 2 points}) = P(WW) \\ = 0.5 \times 0.3 \\ = 0.15.$$

$$(b) \quad P(X \text{ scores less than } 1\frac{1}{2} \text{ points}) \\ = 1 - P(1\frac{1}{2} \text{ points}) - P(2 \text{ points}) \\ = 1 - P(WD + DW) - P(WW) \\ = 1 - (0.5 \times 0.2) - (0.2 \times 0.3) - 0.15 \\ = 1 - 0.1 - 0.06 - 0.15 \\ = 0.69.$$



Probability BOS

Part A: Introduction to Probability

Question 2

1981

“In a certain State, of population 4.8 million, about 1 200 persons are killed on the roads each year. Therefore, there is an approximately 1 in 4 000 chance of a particular person being killed in a road accident each year.”

Is this statement valid? Give brief reasons for your answer.

This statement assumes that each person is equally likely to be killed in a road accident. This is incorrect because some people travel on the roads more than others, some are better drivers, while others may not use the roads at all. Hence the statement is not valid.

Question 3

1997

Forty-five balls, numbered 1 to 45, are placed in a barrel, and one ball is drawn at random. What is the probability that the number on the ball drawn is even?

2

$$P(\text{Even}) = \frac{22}{45}.$$

Question 4

2016

In a raffle, 30 tickets are sold and there is one prize to be won.

What is the probability that someone buying 6 tickets wins the prize?

(A) $\frac{1}{30}$

(B) $\frac{1}{6}$

(C) $\frac{1}{5}$

(D) $\frac{1}{4}$



Probability BOS

Part A: Introduction to Probability

Question 5

1994

'A bag contains green, black, and red jellybeans. Therefore, if I choose one jellybean at random from the bag, the probability that it is black is $\frac{1}{3}$.'

Is this statement true or false? Explain why, in no more than one sentence.

The statement is false unless there are equal numbers of green, black and red jelly beans.

Question 6

1995

A factory assembles torches. Each torch requires one battery and one bulb. It is known that 6% of all batteries and 4% of all bulbs are defective.

2

Find the probability that, in a torch selected at random, both the battery and the bulb are NOT defective. Give your answer in exact form.

$$\begin{aligned} P(\text{Both not defective}) &= 0.94 \times 0.96 \\ &= 0.9024 \text{ (by calc.)} \end{aligned}$$

Part A: Introduction to Probability

Question 7

2013

A bag contains 4 red marbles and 6 blue marbles. Three marbles are selected at random without replacement.

What is the probability that at least one of the marbles selected is red?

(A) $\frac{1}{6}$

(B) $\frac{1}{2}$

(C) $\frac{5}{6}$

(D) $\frac{29}{30}$

Question 8

2014

Three runners compete in a race. The probabilities that the three runners finish the race in under 10 seconds are $\frac{1}{4}$, $\frac{1}{6}$ and $\frac{2}{5}$ respectively.

What is the probability that at least one of the three runners will finish the race in under 10 seconds?

(A) $\frac{1}{60}$

(C) $\frac{3}{8}$

(B) $\frac{37}{60}$

(D) $\frac{5}{8}$

$$P(\text{none finishes in 10 seconds}) = \frac{3}{4} \times \frac{5}{6} \times \frac{3}{5} = \frac{3}{8}$$

$$P(\text{at least one will finish in 10 seconds}) = 1 - \frac{3}{8} = \frac{5}{8}$$

Answer: D



Question 1

1977

In a certain school the senior students are divided into two classes A and B . Class A contains 11 female and 9 male students and class B contains 8 female and 12 male students. A certain occasion requires two speakers, one from each class. These are selected at random. Find the probabilities that:

- (a) both speakers are male;
- (b) both speakers are female;
- (c) one speaker is male and one is female.

$$(a) P(MM) = \frac{9}{20} \times \frac{12}{20} = \frac{108}{400} = \frac{27}{100}.$$

$$(b) P(FF) = \frac{11}{20} \times \frac{8}{20} = \frac{88}{400} = \frac{11}{50}.$$

$$\begin{aligned} (c) P &= P(MF) + P(FM) \\ &= \frac{9}{20} \times \frac{8}{20} + \frac{11}{20} \times \frac{12}{20} \\ &= \frac{72}{400} + \frac{132}{400} \\ &= \frac{51}{100}. \end{aligned}$$

Question 2

1978

An urn contains 4 black and 3 white balls. Two balls are drawn at random and placed in a hat. What is the probability that the hat contains

- (a) two white balls;
- (b) a white and a black ball?

$$(a) P = \frac{3}{7} \times \frac{2}{6} = \frac{6}{42} = \frac{1}{7}.$$

$$\begin{aligned} (b) P &= P(WB) + P(BW) \\ &= \frac{3}{7} \times \frac{4}{6} + \frac{4}{7} \times \frac{3}{6} \\ &= \frac{12}{42} + \frac{12}{42} = \frac{24}{42} \\ &= \frac{4}{7}. \end{aligned}$$



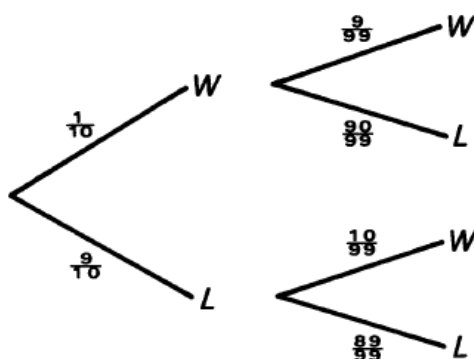
Question 3

1976

One hundred tickets are to be sold in a raffle. Two different tickets are to be drawn out for first and second prizes respectively. A man buys ten tickets. Find the probabilities that:

- he wins the first prize;
- he wins both prizes;
- he wins neither prize;
- he wins at least one prize.

(ii)



$$(a) \quad P = \frac{10}{100} = \frac{1}{10}.$$

$$(b) \quad P(WW) = \frac{10}{100} \times \frac{9}{99} = \frac{1}{110}.$$

$$(c) \quad P(LL) = \frac{90}{100} \times \frac{89}{99} = \frac{89}{110}.$$

$$\begin{aligned} (d) \quad P(\text{at least 1 prize}) &= 1 - P(LL) \\ &= 1 - \frac{89}{110} \\ &= \frac{21}{110}. \end{aligned}$$



Question 4

1980

(ii) A box contains 5 good and 3 defective light bulbs. Two are drawn at random.

(a) What is the probability that the first one drawn is defective?

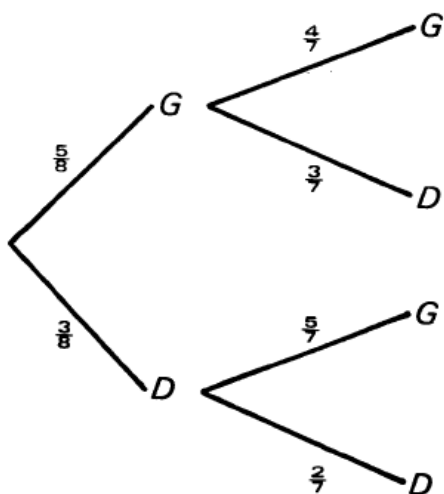
By drawing a tree diagram, or otherwise, calculate the probability that the two light bulbs drawn are:

(b) both defective;

(c) both good;

(d) one defective and one good.

(ii)



$$(a) P = \frac{3}{8}$$

$$(b) P(DD) = \frac{3}{8} \times \frac{2}{7} = \frac{6}{56} = \frac{3}{28}$$

$$(c) P(GG) = \frac{5}{8} \times \frac{4}{7} = \frac{20}{56} = \frac{5}{14}$$

$$\begin{aligned} (d) P &= 1 - (P(DD) + P(GG)) \\ &= 1 - \left(\frac{3}{28} + \frac{10}{28} \right) \\ &= 1 - \frac{13}{28} \\ &= \frac{15}{28} \end{aligned}$$



Question 5

1982

Five similar discs are taken and each marked with one and only one letter. Two are marked with *A*, one with *H*, one with *L* and the final one with *O*. They are then placed in a box. The five discs are drawn at random one at a time from the box without replacement.

What is the probability that:

- (a) the first disc drawn is marked *A*?
- (b) the order of selection of the five discs spells the word *ALOHA*?

$$(a) P(\text{First disc } A) = \frac{2}{5}.$$

$$\begin{aligned} (b) P(ALOHA) &= \frac{2}{5} \times \frac{1}{4} \times \frac{1}{3} \times \frac{1}{2} \times \frac{1}{1} \\ &= \frac{2}{120} \\ &= \frac{1}{60}. \end{aligned}$$

Question 6

1983

There are two prizes in a raffle in which 53 tickets are sold. The first prize is obtained by drawing a ticket at random, and this ticket is not replaced for the draw for the second prize. A man buys two tickets in the raffle. What is the probability that:

- (a) he wins the first prize;
- (b) he wins both prizes;
- (c) he does not win a prize?

$$(a) P(\text{first prize}) = \frac{2}{53}.$$

$$\begin{aligned} (b) P(\text{both prizes}) &= \frac{2}{53} \times \frac{1}{52} \\ &= \frac{1}{1378}. \end{aligned}$$

$$\begin{aligned} (c) P(\text{no prizes}) &= \frac{51}{53} \times \frac{50}{52} \\ &= \frac{1275}{1378}. \end{aligned}$$



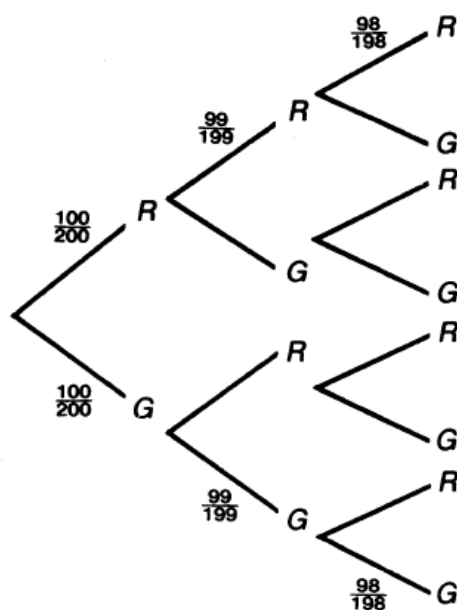
Question 7

1985

100 green tickets and 100 red tickets are sold in a raffle in which there are three prizes. One ticket is drawn for first prize, discarded, and then a second ticket is drawn for the second prize. This is also discarded before the third prize is drawn.

- (a) What is the probability that all three prizes are won by green tickets?
- (b) What is the probability that at least one green ticket wins a prize?

(iii) (a)



$$P(GGG) = \frac{100}{200} \times \frac{99}{199} \times \frac{98}{198}$$

$$= \frac{49}{398}.$$

(b) $P(\text{at least 1 green ticket})$

$$= 1 - P(RRR)$$

$$= 1 - \frac{49}{398} \text{ (N.B. } P(RRR) = P(GGG))$$

$$= \frac{349}{398}.$$

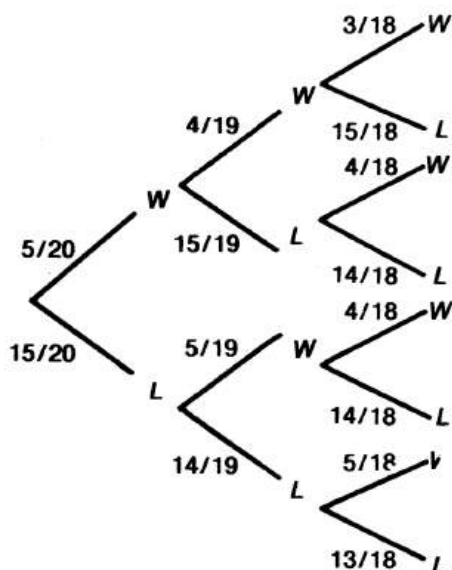


Question 8

1987

Michele buys five tickets in a raffle in which 20 tickets are sold. Three different tickets are to be drawn out for first, second and third prizes. Using a tree diagram find the probability that:

- (a) Michele wins all three prizes;
- (b) Michele does not win a prize;
- (c) Michele wins at least one prize;
- (d) Michele wins exactly one prize.



$$(a) P(WWW) = \frac{5}{20} \times \frac{4}{19} \times \frac{3}{18} = \frac{1}{114}.$$

$$(b) P(LLL) = \frac{15}{20} \times \frac{14}{19} \times \frac{13}{18} = \frac{91}{228}.$$

$$(c) P(\text{at least one prize}) = 1 - P(LLL) = 1 - \frac{91}{228} = \frac{137}{228}.$$

$$(d) P(\text{exactly 1 prize}) = P(WLL) + P(LWL) + P(LLW)$$

$$P(WLL) = \frac{5}{20} \times \frac{15}{19} \times \frac{14}{18} = \frac{35}{228}$$

$$P(LWL) = \frac{15}{20} \times \frac{5}{19} \times \frac{14}{18} = \frac{35}{228}$$

$$P(LLW) = \frac{15}{20} \times \frac{14}{19} \times \frac{5}{18} = \frac{35}{228}$$

$$\therefore P(\text{exactly 1 prize})$$

$$= \frac{35}{228} + \frac{35}{228} + \frac{35}{228}$$

$$= \frac{105}{228}$$

$$= \frac{35}{76}.$$



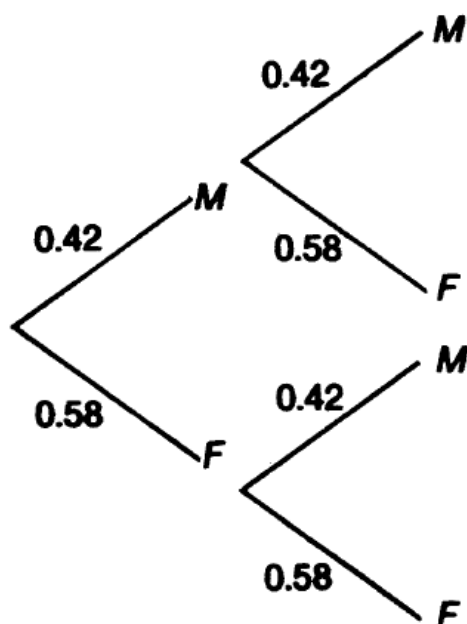
Question 9

1989

In a large school, the student population is 42% male and 58% female. Two students are selected at random to take part in a survey.

Find, correct to two decimal places, the probability that :

- (i) both are female;
- (ii) both are of the same sex;
- (iii) they are of different sexes.



$$\begin{aligned}
 \text{(i) } P(FF) &= 0.58 \times 0.58 \\
 &= 0.3364 \quad (\text{by calc.}) \\
 &= 0.34 \text{ to 2 decimal places.}
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii) } P(FF \text{ or } MM) \\
 &= 0.58 \times 0.58 + 0.42 \times 0.42 \\
 &= 0.5128 \quad (\text{by calc.}) \\
 &= 0.51 \text{ to 2 decimal places.}
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii) } P(MF \text{ or } FM) \\
 &= 0.42 \times 0.58 + 0.58 \times 0.42 \\
 &= 0.4872 \\
 &= 0.49 \text{ to 2 decimal places.}
 \end{aligned}$$



Question 10

1990

A box contains 8 red and 11 green marbles. Sue-Mei randomly selects three marbles one at a time and without replacement. What is the probability that she selects

green, red, green

in that order ?

$$\begin{aligned} P(GRG) &= \frac{11}{19} \times \frac{8}{18} \times \frac{10}{17} \\ &= \frac{440}{2907} \end{aligned}$$

Question 11

1991

An insurance company has calculated that the probability of a woman being alive in 40 years time is 0.8 and that the probability of her husband being alive in 40 years time is 0.7. What is the probability that in 40 years time:

- (i) both will be alive;
- (ii) only one of them will be alive?

$$\begin{aligned} P(W_{\text{alive}}) &= 0.8 \\ \therefore P(W_{\text{dead}}) &= 1 - 0.8 \\ &= 0.2 \\ P(H_{\text{alive}}) &= 0.7 \\ \therefore P(H_{\text{dead}}) &= 1 - 0.7 \\ &= 0.3 \end{aligned}$$

$$\begin{aligned} \text{(i) } P(\text{both alive}) &= P(W_{\text{alive}} H_{\text{alive}}) \\ &= 0.8 \times 0.7 \\ &= 0.56. \end{aligned}$$

$$\begin{aligned} \text{(ii) } P(\text{only 1 alive}) &= P(W_{\text{alive}} H_{\text{dead}} \text{ or } W_{\text{dead}} H_{\text{alive}}) \\ &= 0.8 \times 0.3 + 0.2 \times 0.7 \\ &= 0.24 + 0.14 \\ &= 0.38. \end{aligned}$$

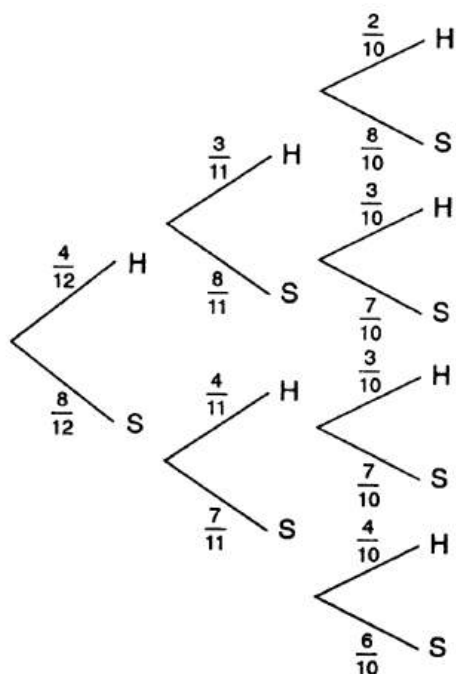


Question 12

1992

A box contains twelve chocolates all of exactly the same appearance. Four of the chocolates are hard and eight are soft. Kim eats three chocolates chosen randomly from the box. Using a tree diagram, or otherwise, find the probability that:

- the first chocolate Kim eats is hard;
- Kim eats three hard chocolates;
- Kim eats exactly one hard chocolate.



$$(i) \quad P(H) = \frac{4}{12} = \frac{1}{3}.$$

$$(ii) \quad P(HHH) = \frac{4}{12} \times \frac{3}{11} \times \frac{2}{10} \\ = \frac{1}{55}.$$

$$(iii) \quad P(\text{exactly 1 hard chocolate}) \\ = P(HSS) + P(SHS) + P(SSH) \\ = \left(\frac{4}{12} \times \frac{8}{11} \times \frac{7}{10}\right) + \left(\frac{8}{12} \times \frac{4}{11} \times \frac{7}{10}\right) \\ \quad \quad \quad + \left(\frac{8}{12} \times \frac{7}{11} \times \frac{4}{10}\right) \\ = \frac{28}{165} + \frac{28}{165} + \frac{28}{165} \\ = \frac{84}{165}.$$



Question 13

1993

The die used in a new game has 20 faces. Each face has a different letter of the alphabet marked on it, however the letters Q , U , V , X , Y , and Z have not been used.

- (i) The die is rolled twice. What is the probability that the same letter appears on the upper face twice?
- (ii) The die is rolled three times. What is the probability that the letter E appears on the upper face exactly twice?

(i) Prob. of rolling a particular letter twice

$$= \frac{1}{20} \times \frac{1}{20}$$

$$= \frac{1}{400}$$

Prob. of rolling any of 20 letters twice

$$= \frac{1}{400} \times 20$$

$$= \frac{20}{400}$$

$$= \frac{1}{20}.$$

(ii) $P(E \text{ appearing twice})$

$$= P(EE\tilde{E}) + P(E\tilde{E}E) + P(\tilde{E}EE)$$

$$= \left(\frac{1}{20} \times \frac{1}{20} \times \frac{19}{20}\right) + \left(\frac{1}{20} \times \frac{19}{20} \times \frac{1}{20}\right) + \left(\frac{19}{20} \times \frac{1}{20} \times \frac{1}{20}\right)$$

$$= \frac{19}{8000} + \frac{19}{8000} + \frac{19}{8000}$$

$$= \frac{57}{8000}.$$



Question 14

1995

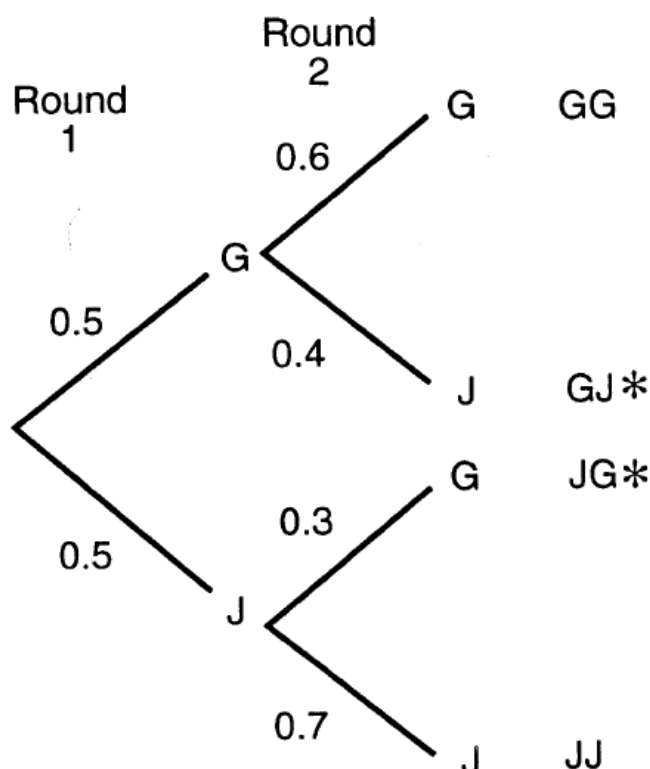
Greg and Jack are playing in a golf tournament. They will play two rounds and each has an equal chance of winning the first round. 4

If Greg wins the first round, his probability of winning the second round is increased to 0.6.

If Greg loses the first round, his probability of winning the second round is reduced to 0.3.

- (i) Draw a tree diagram for the two-round sequence. Label each branch of the diagram with the appropriate probability.
- (ii) Find the probability that Greg wins exactly one round.

(i)



$$\begin{aligned}
 \text{(ii) } P(\text{Greg wins one round}) &= P(GJ) + P(JG) \\
 &= 0.5 \times 0.4 + 0.5 \times 0.3 \\
 &= 0.2 + 0.15 \\
 &= 0.35.
 \end{aligned}$$



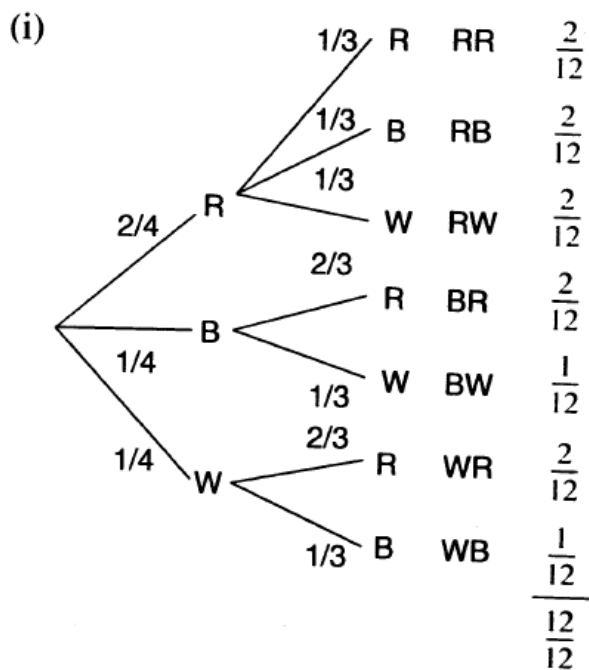
Question 15

1997

5

A bag contains two red balls, one black ball, and one white ball. Andrew selects one ball from the bag and keeps it hidden. He then selects a second ball, also keeping it hidden.

- Draw a tree diagram to show all the possible outcomes.
- Find the probability that both the selected balls are red.
- Find the probability that at least one of the selected balls is red.
- Andrew drops one of the selected balls and we can see that it is red. What is the probability that the ball that is still hidden is also red?



$$(ii) P(RR) = \frac{2}{12}$$

$$= \frac{1}{6}$$

$$(iii) P(\text{At least 1 red}) = 1 - [P(BW) + P(WB)]$$

$$= 1 - \left(\frac{1}{12} + \frac{1}{12}\right)$$

$$= 1 - \frac{2}{12}$$

$$= \frac{10}{12}$$

$$= \frac{5}{6}$$

- (iv) If one of the selected balls is red, there are 5 possible outcomes only one of which will yield another red:

RR, RB, RW, BR and WR

$$\therefore P(\text{Hidden Red}) = \frac{1}{5}$$



Question 16

1999

A box contains five cards. Each card is labelled with a number. The numbers on the cards are 0, 3, 3, 5, 5.

5

Cameron draws one card at random from the box and then draws a second card at random *without* replacing the first card drawn.

- (i) What is the probability that he draws a '5', then a '3'?
- (ii) What is the probability that the sum of the two numbers drawn is at least 8?
- (iii) What is the probability that the second card drawn is labelled '3'?

$$\begin{aligned}
 \text{(b) (i) } P(5 \text{ then } 3) &= P(5) \times P(3) \\
 &= \frac{2}{5} \times \frac{2}{4} \\
 &= \frac{1}{5}.
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii) } P(\text{sum at least } 8) &= P(\text{sum } 8) + P(\text{sum } 10) \\
 &= P(5, 3 \text{ or } 3, 5) + P(5, 5) \\
 &= \frac{1}{5} + \frac{1}{5} + \frac{2}{5} \times \frac{1}{4} \\
 &= \frac{1}{2}.
 \end{aligned}$$

- (iii) The probability that the second card is a 3 is the same as the probability that the first card is a 3, that is, $\frac{2}{5}$.

$$\begin{aligned}
 \text{OR } P(x, 3) &= P(0, 3) + P(3, 3) + P(5, 3) \\
 &= \frac{1}{5} \times \frac{2}{4} + \frac{2}{5} \times \frac{1}{4} + \frac{2}{5} \times \frac{2}{4} \\
 &= \frac{2}{5}.
 \end{aligned}$$

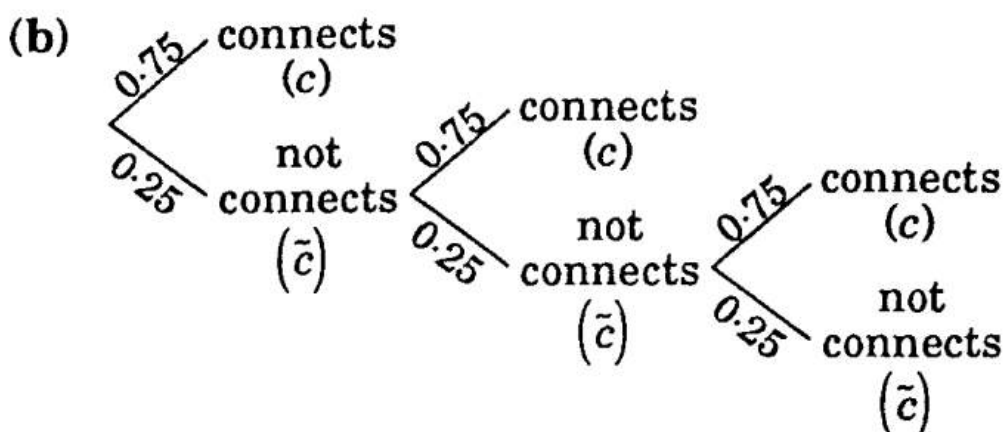


Question 17

2001

Onslo tries to connect to his internet service provider. The probability that he connects on any single attempt is 0.75.

- (i) What is the probability that he connects for the first time on his second attempt? 2
- (ii) What is the probability that he is still not connected after his third attempt? 1



$$(i) \quad P(\tilde{c}c) = 0.25 \times 0.75 \\ = 0.1875.$$

$$(ii) \quad P(\tilde{c}\tilde{c}\tilde{c}) = 0.25 \times 0.25 \times 0.25 \\ = 0.015\,625.$$



Question 18

2001

Five candidates, A , B , C , D and E , are standing for an election. Their names are written on pieces of cardboard that are placed in a barrel and are drawn out randomly to determine their positions on the ballot paper.

- (i) What is the probability that A is drawn first? 1
- (ii) What is the probability that the order of the names on the ballot paper is that shown below? 2



(b) (i) $P(A \text{ drawn first}) = \frac{1}{5}.$

(ii) There are $5 \times 4 \times 3 \times 2 \times 1 = 120$ possible arrangements,

$$\therefore P(\text{one particular arrangement}) = \frac{1}{120}.$$

OR $P(A \text{ then } B \text{ then } C \text{ then } D \text{ then } E)$

$$= P(A \text{ 1st}) \times P(B \text{ 2nd}) \times P(C \text{ 3rd}) \\ \times P(D \text{ 4th}) \times P(E \text{ 5th})$$

$$= \frac{1}{5} \times \frac{1}{4} \times \frac{1}{3} \times \frac{1}{2} \times 1$$

$$= \frac{1}{120}.$$



Question 19

2002

Chris has four pairs of socks in a drawer, each pair a different colour.

He selects socks one at a time and at random from the drawer.

- (i) The probability that he does NOT have a matching pair after selecting the second sock is $\frac{6}{7}$. Explain why this is so. 1
- (ii) Find the probability that he does NOT have a matching pair after selecting the third sock. 2
- (iii) What is the probability that the first three socks include a matching pair? 1

- (c) (i) The first sock selected can be any colour.
The second sock must not be the same colour as the first drawn.

$$\begin{aligned}\therefore \text{Probability} &= \frac{8}{8} \times \frac{6}{7} \\ &= \frac{6}{7}.\end{aligned}$$

- (ii) The third sock cannot be the same colour as the first two colours drawn.

$$\begin{aligned}\therefore \text{Required probability} &= \frac{8}{8} \times \frac{6}{7} \times \frac{4}{6} \\ &= \frac{4}{7}.\end{aligned}$$

$$\begin{aligned}\text{(iii) } P(\text{matching pair}) &= 1 - \frac{4}{7} \\ &= \frac{3}{7}.\end{aligned}$$

Probability BOS

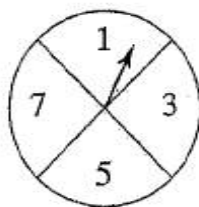
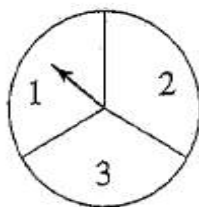
Part B: Tree Diagram



Question 20

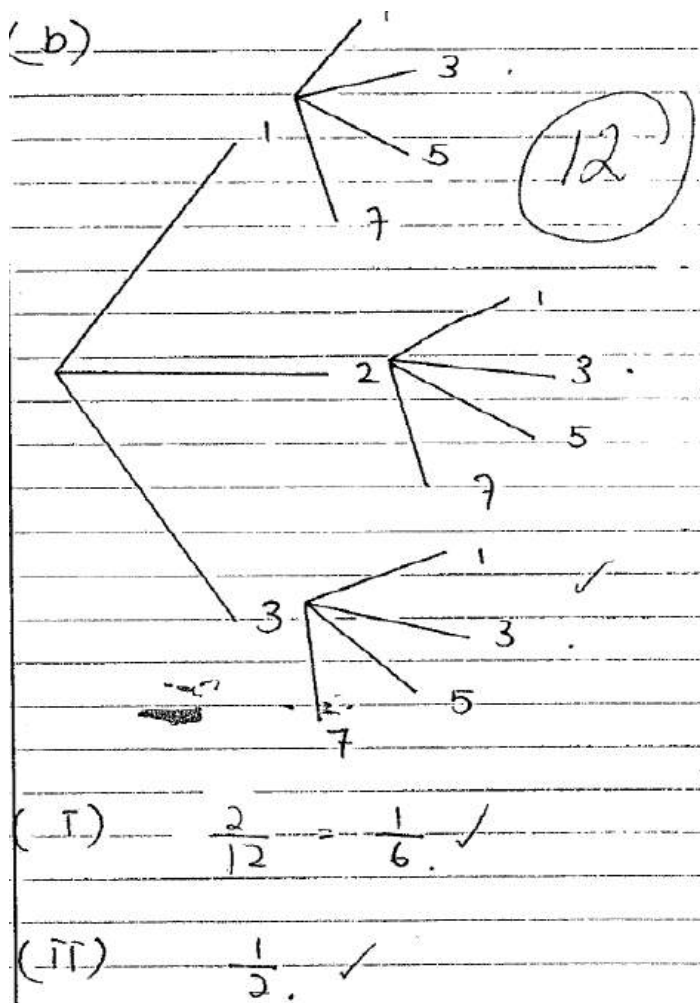
2003

The diagram shows two spinners which are spun simultaneously.



Each of the three outcomes on the first spinner are equally likely, and each of the four outcomes on the second spinner are equally likely.

- (i) What is the probability that both spinners stop on the same number? 2
- (ii) What is the probability that at least one of the spinners stops on a 3? 1





Question 21

2005

A total of 300 tickets are sold in a raffle which has three prizes. There are 100 red, 100 green and 100 blue tickets.

At the drawing of the raffle, winning tickets are NOT replaced before the next draw.

- (i) What is the probability that each of the three winning tickets is red? 2
- (ii) What is the probability that at least one of the winning tickets is not red? 1
- (iii) What is the probability that there is one winning ticket of each colour? 2

$$(d) (i) P(RRR) = \frac{100}{300} \times \frac{99}{299} \times \frac{98}{298}$$

Note: The number of tickets decreases with each draw.

$$P(RRR) = \frac{970\,200}{26\,730\,600} = \frac{1617}{44\,551}$$

$$(ii) P(\text{at least one is not red})$$

$$= 1 - P(RRR)$$

$$= 1 - \frac{1617}{44\,551}$$

$$= \frac{42\,934}{44\,551}$$

$$(= 0.963\,704\,518 \dots)$$

$$(= 0.964 \text{ to 3 dec. places. })$$

$$(iii) P(\text{one of each colour})$$

$$= P(RGB) + P(RBG) + P(BRG) + P(BGR)$$

$$+ P(GRB) + P(GBR)$$

$$= 6 \times \left[\frac{100}{300} \times \frac{100}{299} \times \frac{100}{298} \right]$$

$$= \frac{20\,000}{89\,102}$$

$$= \frac{10\,000}{44\,551}$$

$$= \frac{10\,000}{44\,551}$$

$$= 0.224\,461\,852 \dots$$

$$(= 0.224 \text{ to 3 dec. places. })$$



Question 22

2006

A chessboard has 32 black squares and 32 white squares. Tanya chooses three different squares at random.

- | | |
|---|---|
| (i) What is the probability that Tanya chooses three white squares? | 2 |
| (ii) What is the probability that the three squares Tanya chooses are the same colour? | 1 |
| (iii) What is the probability that the three squares Tanya chooses are not the same colour? | 1 |

$$\begin{aligned} \text{(c) (i) } P(\text{white, white, white}) &= \frac{32}{64} \times \frac{31}{63} \times \frac{30}{62} \\ &= \frac{5}{42} \text{ or } 0.119... \end{aligned}$$

$$\begin{aligned} \text{(ii) } P(\text{same colour}) &= P(\text{white, white, white}) \\ &\quad + P(\text{black, black, black}) \\ &= \frac{32}{64} \times \frac{31}{63} \times \frac{30}{62} \\ &\quad + \frac{32}{64} \times \frac{31}{63} \times \frac{30}{62} \\ &= \frac{5}{42} + \frac{5}{42} \\ &= \frac{5}{21} \end{aligned}$$

$$\begin{aligned} \text{(iii) } P(\text{not the same colour}) &= 1 - P(\text{same colour}) \\ &= 1 - \frac{5}{21} \\ &= \frac{16}{21} \end{aligned}$$



Question 23

2009

On each working day James parks his car in a parking station which has three levels. He parks his car on a randomly chosen level. He always forgets where he has parked, so when he leaves work he chooses a level at random and searches for his car. If his car is not on that level, he chooses a different level and continues in this way until he finds his car.

- (i) What is the probability that his car is on the first level he searches? 1
- (ii) What is the probability that he must search all three levels before he finds his car? 1
- (iii) What is the probability that on every one of the five working days in a week, his car is not on the first level he searches? 1

$$\begin{aligned} \text{(i)} \quad & \frac{1}{3} \\ \text{(ii)} \quad & \frac{2}{3} \times \frac{1}{2} \times 1 = \frac{1}{3} \\ \text{(iii)} \quad & \left(\frac{2}{3}\right)^5 = \frac{32}{243} \end{aligned}$$

Question 24

2010

There are twelve chocolates in a box. Four of the chocolates have mint centres, four have caramel centres and four have strawberry centres. Ali randomly selects two chocolates and eats them.

- (i) What is the probability that the two chocolates have mint centres? 1
- (ii) What is the probability that the two chocolates have the same centre? 1
- (iii) What is the probability that the two chocolates have different centres? 1

$$\text{Q4ci } \Pr(\text{both.m.centres}) = \frac{{}^4C_2}{{}^{12}C_2} = \frac{6}{66} = \frac{1}{11}$$

$$\text{Q4cii } \Pr(\text{same.centre}) = 3 \times \frac{1}{11} = \frac{3}{11}$$

$$\text{Q4ciii } \Pr(\text{different.centres}) = 1 - \frac{3}{11} = \frac{8}{11}$$

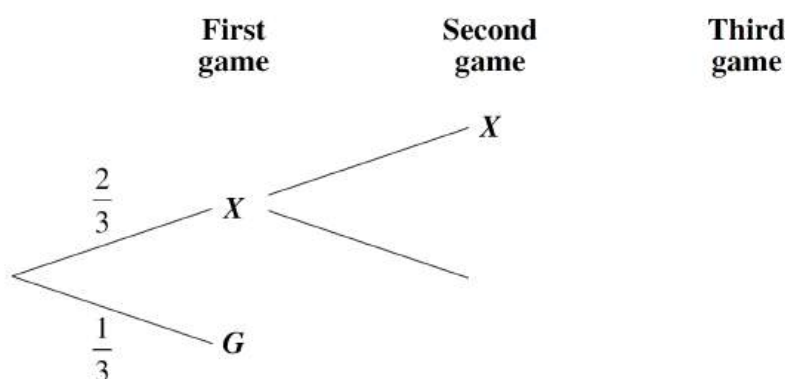


Question 25

2008

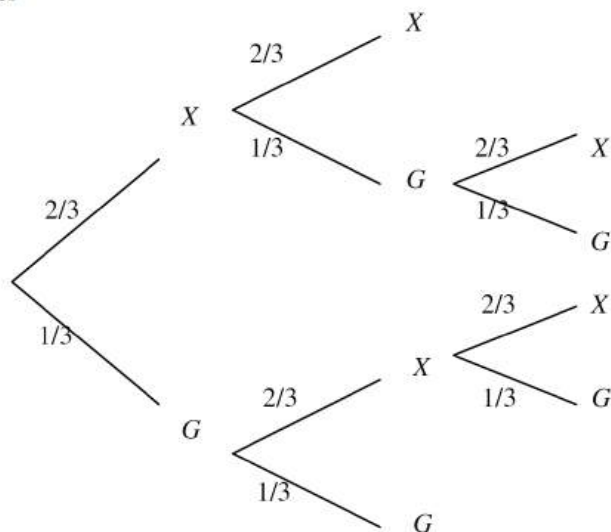
Xena and Gabrielle compete in a series of games. The series finishes when one player has won two games. In any game, the probability that Xena wins is $\frac{2}{3}$ and the probability that Gabrielle wins is $\frac{1}{3}$.

Part of the tree diagram for this series of games is shown.



- Copy and complete the tree diagram showing the possible outcomes. 1
- What is the probability that Gabrielle wins the series? 2
- What is the probability that three games are played in the series? 2

Q7ci



Possible outcomes are $XX, XGX, XGG, GXX, GXG, GG$.

Q7cii $\Pr(XGG) + \Pr(GXG) + \Pr(GG)$
 $= \frac{2}{3} \times \frac{1}{3} \times \frac{1}{3} + \frac{1}{3} \times \frac{2}{3} \times \frac{1}{3} + \frac{1}{3} \times \frac{1}{3} = \frac{7}{27}$

Q7ciii $\Pr(3 \text{ games}) = 1 - \Pr(2 \text{ games}) = 1 - \left(\left(\frac{1}{3} \right)^2 + \left(\frac{2}{3} \right)^2 \right) = \frac{4}{9}$



Question 26

2011

Kim has three red shirts and two yellow shirts. On each of the three days, Monday, Tuesday and Wednesday, she selects one shirt at random to wear. Kim wears each shirt that she selects only once.

- (i) What is the probability that Kim wears a red shirt on Monday? 1
- (ii) What is the probability that Kim wears a shirt of the same colour on all three days? 1
- (iii) What is the probability that Kim does not wear a shirt of the same colour on consecutive days? 2

$$\text{Q5bi } \Pr(\text{red_shirt_Monday}) = \frac{3}{5}$$

$$\text{Q5bii } \Pr(\text{red_red_red}) = \frac{3}{5} \times \frac{2}{4} \times \frac{1}{3} = \frac{1}{10}$$

$$\begin{aligned} \text{Q5biii } \Pr(\text{red_yellow_red}) + \Pr(\text{yellow_red_yellow}) \\ = \frac{3}{5} \times \frac{2}{4} \times \frac{2}{3} + \frac{2}{5} \times \frac{3}{4} \times \frac{1}{3} = \frac{3}{10} \end{aligned}$$

Question 27

2012

Two buckets each contain red marbles and white marbles. Bucket A contains 3 red and 2 white marbles. Bucket B contains 3 red and 4 white marbles.

Chris randomly chooses one marble from each bucket.

- (i) What is the probability that both marbles are red? 1
- (ii) What is the probability that at least one of the marbles is white? 1
- (iii) What is the probability that both marbles are the same colour? 2

$$\text{Q13ci } \Pr(RR) = \frac{3}{5} \times \frac{3}{7} = \frac{9}{35}$$

$$\text{Q13cii } \Pr(\text{at.least.one.white}) = 1 - \Pr(RR) = 1 - \frac{9}{35} = \frac{26}{35}$$

$$\text{Q13ciii } \Pr(\text{same.colour}) = \Pr(RR) + \Pr(WW) = \frac{9}{35} + \frac{2}{5} \times \frac{4}{7} = \frac{17}{35}$$



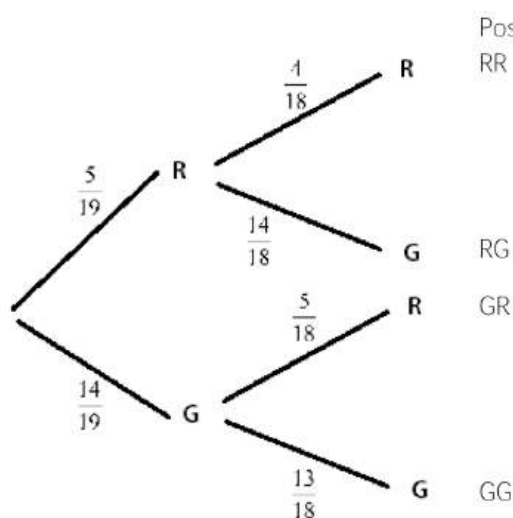
Question 28

2014

A packet of lollies contains 5 red lollies and 14 green lollies. Two lollies are selected at random without replacement.

- (i) Draw a tree diagram to show the possible outcomes. Include the probability on each branch. 2
- (ii) What is the probability that the two lollies are of different colours? 1

Question 12 (c) (i)



Possible Outcomes

RR

Question 12 (c) (ii)

$$P(\text{RG or GR}) = \frac{5}{19} \times \frac{14}{18} + \frac{14}{19} \times \frac{5}{18} = \frac{70}{171}$$

RG

GR

GG

Question 29

2015

The probability that Mel's soccer team wins this weekend is $\frac{5}{7}$.

The probability that Mel's rugby league team wins this weekend is $\frac{2}{3}$.

What is the probability that neither team wins this weekend?

(A) $\frac{2}{21}$

(B) $\frac{10}{21}$

(C) $\frac{13}{21}$

(D) $\frac{19}{21}$

$$P(\text{soccer team loses and rugby team loses}) = \left(1 - \frac{5}{7}\right) \left(1 - \frac{2}{3}\right) = \frac{2}{21}$$

Answer: A



Question 30

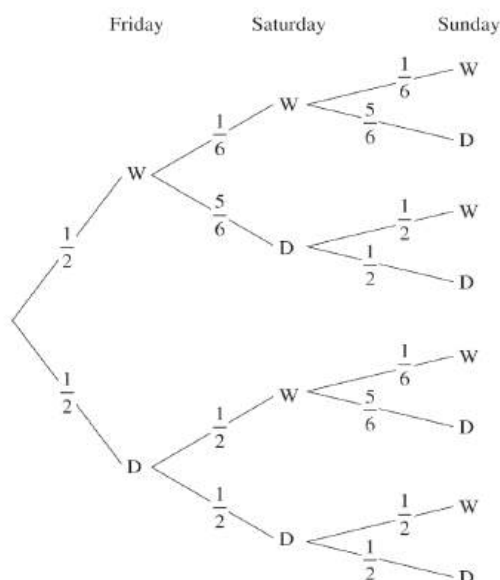
2015

Weather records for a town suggest that:

- if a particular day is wet (W), the probability of the next day being dry is $\frac{5}{6}$
- if a particular day is dry (D), the probability of the next day being dry is $\frac{1}{2}$.

In a specific week Thursday is dry. The tree diagram shows the possible outcomes for the next three days: Friday, Saturday and Sunday.

- (i) Show that the probability of Saturday being dry is $\frac{2}{3}$. 1
- (ii) What is the probability of both Saturday and Sunday being wet? 2
- (iii) What is the probability of at least one of Saturday and Sunday being dry? 1



(b) (i) $P(\text{Saturday is dry}) = P(WD) + P(DD) = \frac{1}{2} \times \frac{5}{6} + \frac{1}{2} \times \frac{1}{2} = \frac{2}{3}$.

(b) (ii) $P(WWW) + P(DWW) = \frac{1}{2} \times \frac{1}{6} \times \frac{1}{6} + \frac{1}{2} \times \frac{1}{2} \times \frac{1}{6} = \frac{1}{18}$

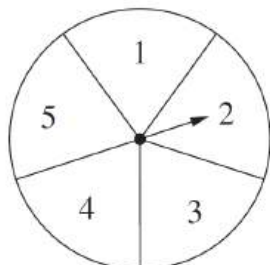
(b) (iii) $1 - [P(WWW) + P(DWW)] = 1 - \frac{1}{18} = \frac{17}{18}$



Question 31

2017

A spinner is marked with the numbers 1, 2, 3, 4 and 5. When it is spun, each of the five numbers is equally likely to occur.



The spinner is spun three times.

- | | |
|---|---|
| (i) What is the probability that an even number occurs on the first spin? | 1 |
| (ii) What is the probability that an even number occurs on at least one of the three spins? | 1 |
| (iii) What is the probability that an even number occurs on the first spin and odd numbers occur on the second and third spins? | 1 |
| (iv) What is the probability that an even number occurs on exactly one of the three spins? | 1 |

Question 12 (e) (i)

Sample answer:

$$P(\text{even number}) = \frac{2}{5}$$

Question 12 (e) (ii)

Sample answer:

Probability (at least 1 even number)

$= 1 - \text{Probability (all odd numbers)}$

$$= 1 - \frac{3}{5} \times \frac{3}{5} \times \frac{3}{5}$$

$$= \frac{98}{125}$$

Question 12 (e) (iii)

Sample answer:

$$P(\text{even, odd, odd}) = \frac{2}{5} \times \frac{3}{5} \times \frac{3}{5}$$

$$= \frac{18}{125}$$

Question 12 (e) (iv)

Sample answer:

$$P(\text{exactly one even}) = 3 \times \frac{2}{5} \times \frac{3}{5} \times \frac{3}{5}$$

$$= \frac{54}{125}$$

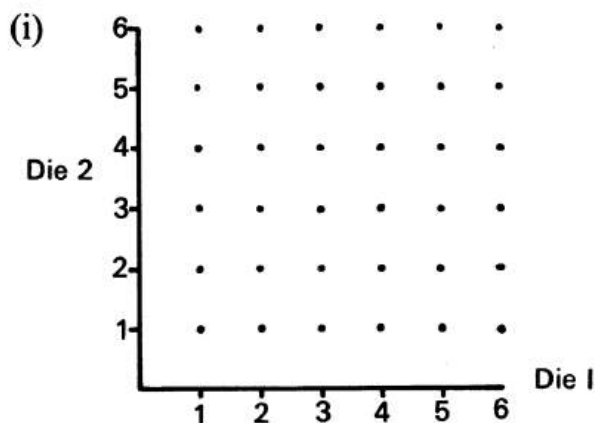


Question 1

1979

Two ordinary dice, with the numbers 1 to 6 on their faces, are thrown. What is the probability that:

- they both show 6?
- they show a 1 and a 6?
- at least one of them shows a 1?
- they show a total of six?



Using a graphical representation to determine the number of favourable outcomes in each case:

- (a) 1 favourable outcome—(6, 6).

$$P = \frac{1}{36} \text{ since } n(s) = 36.$$

- (b) 2 favourable outcomes—(1, 6) and (6, 1).

$$P = \frac{2}{36} = \frac{1}{18}.$$

- (c) 11 favourable outcomes—(1, 1), (1, 2), (2, 1), (1, 3), (3, 1), (1, 4), (4, 1), (1, 5), (5, 1), (1, 6), (6, 1).

$$P = \frac{11}{36}.$$

- (d) 5 favourable outcomes—(1, 5), (2, 4), (3, 3), (4, 2) and (5, 1).

$$P = \frac{5}{36}.$$

Alternatively, the Addition and Product Theorems could be used.

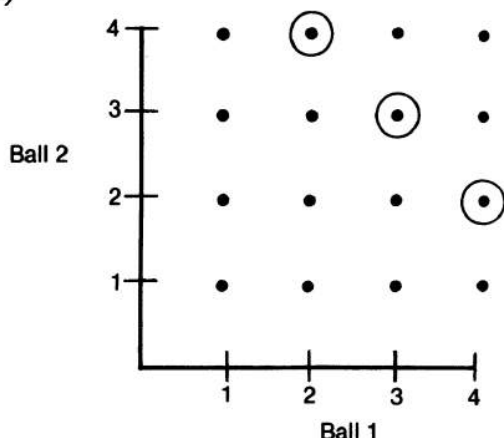


Question 2

1985

The numbers 1, 2, 3, 4 are painted on to four otherwise similar balls. The balls are placed in a bag, and one ball is drawn out at random. The number on this ball is written down and the ball is then replaced in the bag. A second ball is then drawn at random and the number on it written down. Using a tree diagram, or otherwise, what is the probability that the sum of the two written numbers is 6?

(ii)



Using a graphical representation to determine the number of favourable outcomes:

$$P(\text{sum} = 6) = \frac{3}{16}$$

Question 3

1988

- (i) A die whose faces are numbered 1, 2, 3, 4, 5, 6 is tossed twice. The sum S of the numbers which appear uppermost on the die is calculated. Find the probability that S is greater than 8.
- (ii) It is known that a 4 appears on the die at least once in the two throws. Find the probability that S is greater than 8.

(b) (i)

		Die 1					
		1	2	3	4	5	6
Die 2	1	2	3	4	5	6	7
	2	3	4	5	6	7	8
	3	4	5	6	7	8	9
	4	5	6	7	8	9*	10*
	5	6	7	8	9*	10	11
	6	7	8	9	10*	11	12

$$P(S > 8) = \frac{10}{36} \left(\triangle \right) = \frac{5}{18}$$

(ii) Required probability* = $\frac{4}{11}$.



Question 4

1986

Two identical perfect cubes (similar to dice) each having faces numbered 0, 1, 2, 3, 4, 5 are rolled. A score for the roll is determined as the product of the two numbers on the two uppermost faces.

- (a) The cubes are rolled once. What is the probability that the score is
- (α) 0 ?
- (β) at least 16 ?
- (b) If the cubes are rolled twice and the scores for each roll are added, what is the probability of a combined score of at least 41?

(ii) (a)

		Cube 2					
Cube 1		0	1	2	3	4	5
	0	0	0	0	0	0	0
	1	0	1	2	3	4	5
	2	0	2	4	6	8	10
	3	0	3	6	9	12	15
	4	0	4	8	12	16	20
	5	0	5	10	15	20	25

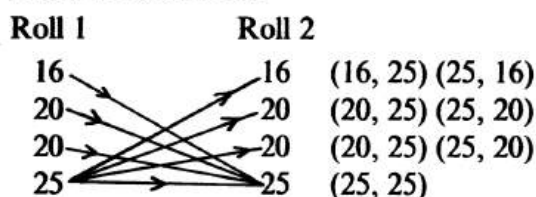
(α) $P(0) = \frac{11}{36}$

(β) $P(\geq 16) = \frac{4}{36} = \frac{1}{9}$

- (b) If the cubes are rolled once, there are 36 outcomes.

If the cubes are rolled twice, there are 36×36 outcomes.

The following combinations give combined scores of at least 41:



$$\therefore P(\geq 41) = \frac{7}{36 \times 36} = \frac{7}{1296}$$



Question 5

1990

Bob tosses two dice with faces numbered 1 to 6. He records the maximum of the two uppermost faces as a score.

- Find the probability that he records the score 1 in a single throw of the two dice.
- Find the probability that he records the scores 1, 1, 1 in three tosses of the two dice.
- Find the probability that he records the score 6 in a single throw of the two dice.

(i)

		Die 1					
		1	2	3	4	5	6
Die 2	1	1	2	3	4	5	6*
	2	2	2	3	4	5	6*
	3	3	3	3	4	5	6*
	4	4	4	4	4	5	6*
	5	5	5	5	5	5	6*
	6	6*	6*	6*	6*	6*	6*

$$P(1) = \frac{1}{36}$$

$$(ii) \quad P(1, 1, 1) = \frac{1}{36} \times \frac{1}{36} \times \frac{1}{36} \\ = \frac{1}{46656}.$$

$$(iii) \quad P(6) = \frac{11}{36} \cdot (*)$$



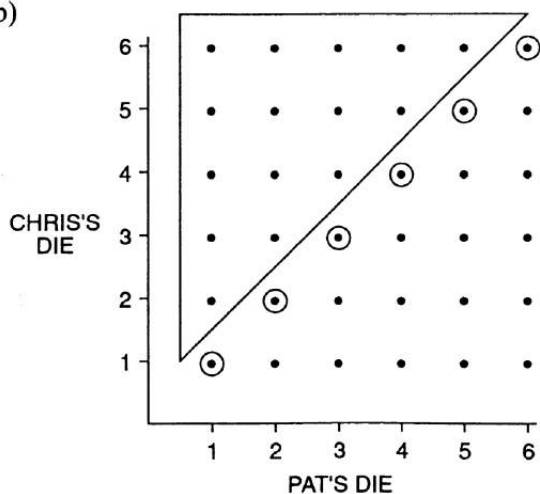
Question 6

1992

Pat and Chris each throw a die.

- Find the probability that they throw the same number.
- Find the probability that the number thrown by Chris is greater than the number thrown by Pat.

(b)



$$(i) \quad P(\text{same number}) = \frac{6}{36} (\odot) = \frac{1}{6}$$

$$(ii) \quad \text{Required probability} = \frac{15}{36} (\triangle) = \frac{5}{12}$$

Question 7

2007

Two ordinary dice are rolled. The score is the sum of the numbers on the top faces.

- What is the probability that the score is 10? 2
- What is the probability that the score is not 10? 1

(b) (i) outcomes are

6	7	8	9	10	11	12
5	6	7	8	9	10	11
4	5	6	7	8	9	10
3	4	5	6	7	8	9
2	3	4	5	6	7	8
1	2	3	4	5	6	7
1	2	3	4	5	6	

$$\therefore P(10) = \frac{3}{36} = \frac{1}{12}$$

$$(ii) P(\text{not } 10) = 1 - P(10) = 1 - \frac{1}{12} = \frac{11}{12}$$



Question 8

1998

5

A game is played in which two coloured dice are thrown once. The six faces of the blue die are numbered 4, 6, 8, 9, 10 and 12. The six faces of the pink die are numbered 2, 3, 5, 7, 11 and 13. The player wins if the number on the pink die is larger than the number on the blue die.

- By drawing up a table of possible outcomes, or otherwise, calculate the probability of the player winning a game.
- Calculate the probability that the player wins at least once in two successive games.

(a) (i)

		PINK DIE					
BLUE		2	3	5	7	11	13
	4			✓	✓	✓	✓
	6				✓	✓	✓
	8					✓	✓
	9					✓	✓
	10					✓	✓
	12						✓

$P(\text{At least one win})$

$$= 1 - P(LL)$$

$$= 1 - \left(\frac{11}{18} \times \frac{11}{18} \right)$$

$$= 1 - \frac{121}{324}$$

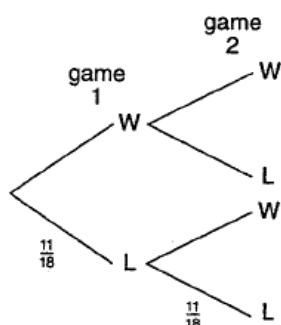
$$= \frac{203}{324}$$

$$P(\text{Win}) = \frac{14}{36}$$

$$= \frac{7}{18}$$

$$(ii) P(\text{Loss}) = 1 - \frac{7}{18}$$

$$= \frac{11}{18}$$





Question 9

2004

In a game, a turn involves rolling two dice, each with faces marked 0, 1, 2, 3, 4 and 5. The score for each turn is calculated by multiplying the two numbers uppermost on the dice.

- (i) What is the probability of scoring zero on the first turn? 2
- (ii) What is the probability of scoring 16 or more on the first turn? 1
- (iii) What is the probability that the sum of the scores in the first two turns is less than 45? 2

(c) Sample space is shown in the table.

$\times$	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	1	2	3	4	5
2	0	2	4	6	8	10
3	0	3	6	9	12	15
4	0	4	8	12	16	20
5	0	5	10	15	20	25

$\therefore$ 36 possible outcomes.

(i) $P(0 \text{ on first turn}) = \frac{11}{36}$.

(ii) $P(16 \text{ or more on first turn}) = \frac{4}{36} = \frac{1}{9}$.

(iii) There are only 3 ways to get a sum of 45 or more on 2 throws:

20 + 25 or 25 + 20 or 25 + 25.

$$\begin{aligned}
 &P(\text{sum} \geq 45) \\
 &= P(20 + 25) + P(25 + 20) + P(25 + 25) \\
 &= \left(\frac{2}{36} \times \frac{1}{36}\right) + \left(\frac{1}{36} \times \frac{2}{36}\right) + \left(\frac{1}{36} \times \frac{1}{36}\right) \\
 &= \frac{5}{1296}.
 \end{aligned}$$

$$\begin{aligned}
 \therefore P(\text{sum} < 45) &= 1 - P(\text{sum} \geq 45) \\
 &= 1 - \frac{5}{1296} \\
 &= \frac{1291}{1296}.
 \end{aligned}$$



Question 1

1980

Six cards labelled A , B , C , D , E and F , are drawn, one at a time, from a hat.

What is the probability that card A or card E will be the third card drawn?

(i) $P(A \text{ is 3rd card drawn})$

$$= \frac{5}{6} \times \frac{4}{5} \times \frac{1}{4} \times \frac{3}{3} \times \frac{2}{2} \times \frac{1}{1}$$

$$= \frac{5}{6} \times \frac{4}{5} \times \frac{1}{4} = \frac{1}{6}$$

Note: If A is the third card drawn, there are:

$5/6$ possibilities for draw 1

(A is not a possibility)

$4/5$ possibilities for draw 2

$1/4$ possibilities for draw 3

(A is the only possibility)

$3/3$ possibilities for draw 4 etc.

Similarly:

$$P(E \text{ is 3rd card drawn}) = \frac{1}{6}$$

$\therefore P(A \text{ or } E \text{ is 3rd card drawn})$

$$= \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3} \text{ since both events are mutually exclusive.}$$



Question 2

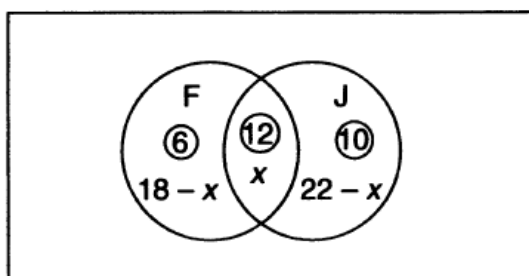
1996

Students studying at least one of the languages, French and Japanese, attend a meeting. Of the 28 students present, 18 study French and 22 study Japanese.

4

- What is the probability that a randomly chosen student studies French?
- What is the probability that two randomly chosen students both study French?
- What is the probability that a randomly chosen student studies both languages?

- (a) Let x be the number of students studying both French and Japanese.



$$\begin{aligned}\text{Now } 18 - x + x + 22 - x &= 28 \\ 40 - x &= 28 \\ x &= 12\end{aligned}$$

$$\begin{aligned}\text{(i) } P(F) &= \frac{18}{28} \\ &= \frac{9}{14}.\end{aligned}$$

$$\begin{aligned}\text{(ii) } P(FF) &= \frac{18}{28} \times \frac{17}{27} \\ &= \frac{17}{42}.\end{aligned}$$

$$\begin{aligned}\text{(iii) } P(\text{Both}) &= \frac{12}{28} \\ &= \frac{3}{7}.\end{aligned}$$